

# TrES-3b

R-0071

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_250502 · created 2026-07-15T16:49:27+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2460797.82282 +/- 0.00081 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1841 +/- 0.0093                 |
| Transit depth (Rp/Rs)^2                           | 3.39 +/- 0.34 [%]                 |
| Orbital Inclination (inc)                         | 84.05 +/- 0.65                    |
| Airmass coefficient 1 (a1)                        | 1.048 +/- 0.012                   |
| Airmass coefficient 2 (a2)                        | -0.0373 +/- 0.0067                |
| Scatter in the residuals of the lightcurve fit is | 1.11 %                            |
| Best Comparison Star                              | #2 - [601, 418]                   |
| Optimal Aperture                                  | 4.14                              |
| Optimal Annulus                                   | 13.56                             |
| Transit Duration (day)                            | 0.0721 +/- 0.0036                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                           |
|--------------------------------------|-----------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.1841 $\pm$ 0.0093 vs 0.16309 (deviation 2.26 $\sigma$ ) |
| Duration fit vs published            | 0.0721 d vs 0.0567 d (deviation 4.29 $\sigma$ )           |
| Mid-time O-C                         | -5.6 min                                                  |
| Depth SNR                            | 19.8                                                      |
| Residual scatter                     | 1.11 %                                                    |

### Light curve

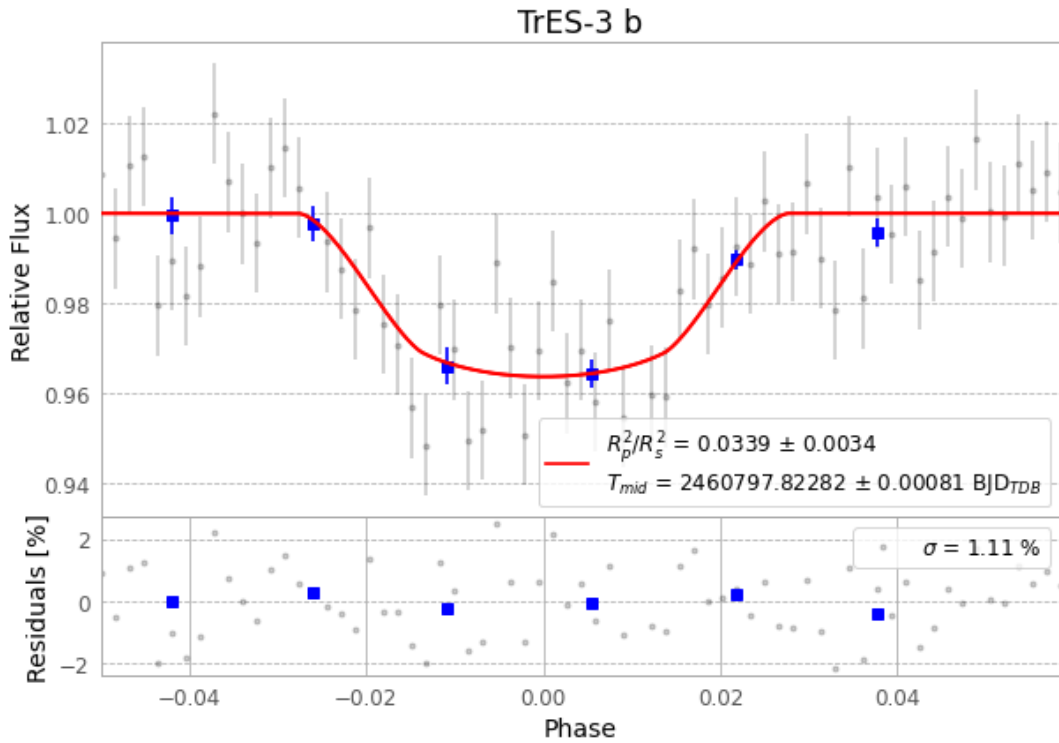


Figure 1 — FinalLightCurve\_TrES-3 b\_2025-05-01.png (SHA-256 90a757f074002b66...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 202], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 4c8de1f9481ade5b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

## Data provenance

69 FITS frames (45 MB), TRES-3250502060616.FITS → TRES-3250502093015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-250502.log (053010a49959...), FinalLightCurve\_TrES-3 b\_2025-05-01.png (90a757f07400...), FinalLightCurve\_TrES-3 b\_2025-05-01.csv (8116b8cedb55...)

## Compute resources

Wall time 145.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:49Z.